



**Davao del Norte State College
Institute of Computing**

Bachelor of Science in Information Systems

IS312 – ENTERPRISE ARCHITECTURE

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I. Plastic Flamingo (PLAF)

Enterprise Overview

Plastic Flamingo (PLAF) is a non-profit organization located in the Philippines that was established in 2018 with the aim of solving the problem of plastic pollution. The company's goal is eco-lumber. is PLAF to collaborate, collect with, and local recycle governments, the community's plastic and waste businesses, and transform them in order to achieve a sustainable circular product economy that addresses the challenges of plastic waste while optimizing resource utilization and sustainability. In this way, PLAF has become the number one icon in creating the zero-waste culture. It will also help in saving environmental the activism marine with ecosystems the ad notion decreases of our social reliance responsibility. It raises resource awareness [1].

Local PLAF management, along with the population, plays a significant role in providing employment opportunities within the company. This collaboration not only helps in effectively managing company waste but also contributes to achieving global sustainable development goals. The organization operates in various locations and has a vision to expand its efforts in Southeast Asia and beyond. By doing so, they aim to enhance their impact on sustainable development globally, promoting environmental stewardship and socio-economic growth in more regions.

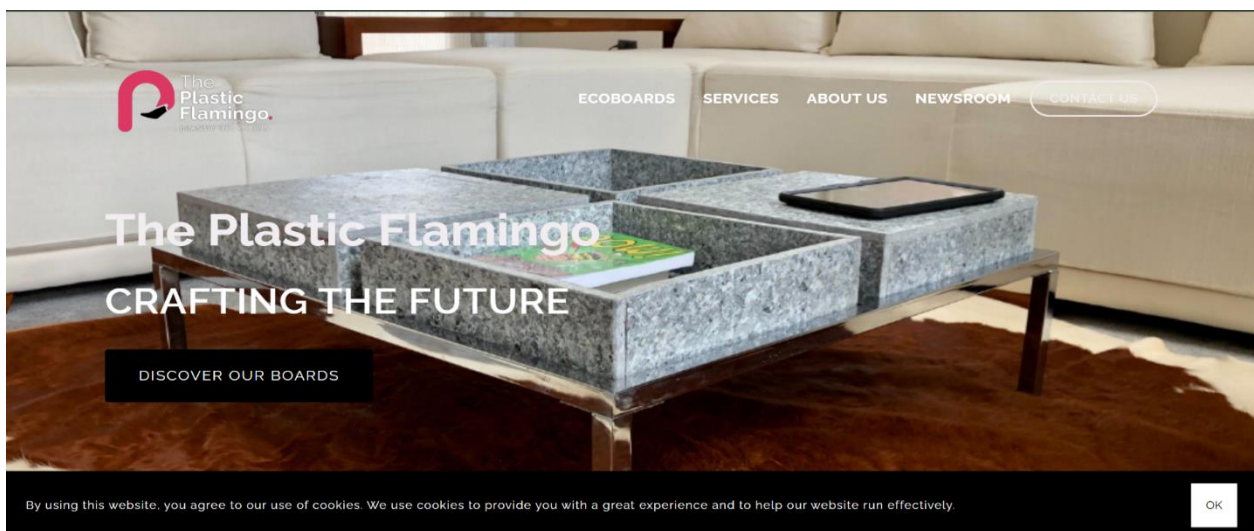


Figure 1. The Plastic Flamingo Website

Goals and Initiatives

Plastic Flamingo (PLAF's) primary goal is to address the environmental and social issues caused by plastic waste by transforming it into valuable resources. To achieve this, the organization has formulated several key strategies aimed at maximizing its impact. These strategies include reducing plastic pollution through innovative waste management techniques, promoting sustainable practices to encourage responsible use of resources, and expanding their recycling capacity to handle more plastic waste effectively. Additionally, PLAF aims to scale its global impact by extending its efforts and operations to reach more regions, thereby contributing significantly to global sustainable development goals. Through these initiatives, PLAF is dedicated to fostering a cleaner and more sustainable future.

To support these goals, PLAF has initiated several key projects. They partner with local government units (LGUs) to create efficient plastic collection and sorting programs, ensuring waste is managed effectively. They also run educational campaigns and workshops to raise awareness about proper waste disposal and the benefits of a circular economy. Additionally, PLAF invests in modern recycling technology to boost production and efficiency. They aim to internationalize their efforts by forming partnerships with global organizations to advance zero-waste practices worldwide.

GOALS	INITIATIVES
REDUCE PLASTIC POLLUTION	Partner with LGUs for effective waste management and plastic collection.
PROMOTE SUSTAINABILITY	Educate communities about waste management and circular economies.
EXPAND RECYCLING	Invest in technology to enhance production and processing efficiency.
SCALE GLOBAL IMPACT	Develop partnerships to promote zero-waste cultures internationally.

Table 1: Goals and Initiatives of PLAF

Products and Services

Plastic Flamingo (PLAF) offers innovative products and services aimed at mitigating the environmental impact of plastic waste and promoting sustainability. Their primary product, eco-lumber, is made from recycled plastics and serves as a durable, versatile, and environmentally friendly alternative to traditional wood. It is widely used in construction projects such as outdoor furniture, decking, and fencing, and performs well under various weather conditions, making it a preferred choice for sustainable construction.

In addition to eco-lumber, PLAF provides a range of services to support waste management and sustainability efforts. They collaborate with enterprises and organizations to create custom recycling solutions tailored to specific needs and environmental goals. PLAF also partners with local government units and communities to establish efficient plastic collection programs, ensuring that recyclable materials are processed properly.

PLAF is committed to education and awareness, conducting workshops in schools, businesses, and community groups to teach proper waste disposal practices and the benefits of recycling. They also offer consultancy services to companies looking to improve their waste management strategies and adopt sustainable practices. Through these comprehensive products and services, PLAF strives to make a significant impact on reducing plastic waste and fostering a more sustainable future.

Data and Information

Plastic Flamingo (PLAF) utilizes various types of data to enhance its operations, connect with communities, and make informed decisions.

1. **Community Engagement Data:** This involves tracking participation in workshops and plastic collection activities, along with gathering feedback to shape future plans and assess impact.
2. **Recycling Data:** Detailed records of plastic collection, recycling rates, and eco-lumber production are maintained to monitor operations and ensure quality control.
3. **Supply Chain Data:** Information on partnerships with local government units (LGUs), logistics, and suppliers helps in efficiently managing the flow of materials from collection to recycling.

4. **Employee Data:** Workforce schedules, training records, and performance reviews are monitored to ensure smooth operations and boost employee satisfaction.
5. **Financial Data:** Analysis of revenue, costs, and profit margins helps assess the economic viability of operations and identify areas for cost reduction.
6. **Marketing Data:** Data from campaigns, social media, and outreach events are analyzed to measure the reach and effectiveness of community awareness initiatives.

By leveraging these data types, PLAF improves its operational efficiency, strengthens community connections, and drives sustainable development.

DATA CATEGORY	USAGE
COMMUNITY ENGAGEMENT DATA	Improve strategies and measure impact.
RECYCLING DATA	Track operations and maintain quality.
SUPPLY CHAIN DATA	Manage material flow and logistics.
EMPLOYEE DATA	Organize workforce and training programs.
FINANCIAL DATA	Check sustainability and cut costs.
MARKETING DATA	Measure campaign reach and effectiveness.

Table 2: Data and Information of PLAF

Networks and Infrastructure

Plastic Flamingo (PLAF's) operations are supported by robust networks and infrastructures that ensure efficient and seamless execution of their activities.

1. **Internet Connectivity:** A reliable internet connection is crucial for PLAF's cloud-based systems, enabling real-time communication and coordination with partners and stakeholders.

2. **Local Area Network (LAN):** This on-site network connects various devices such as computers, recycling equipment monitors, and community engagement terminals, ensuring smooth data synchronization and efficient workflow.
3. **Hardware:** Essential equipment includes recycling machinery for processing plastic waste, administrative computers for management tasks, and communication devices to facilitate daily operations.

By leveraging these networks and infrastructures, PLAF effectively manages its processes, enhances communication, and ensures the continuous operation of its recycling and sustainability initiatives.

System and Application

Plastic Flamingo (PLAF) employs various systems and applications to streamline its operations and support its mission:

1. **Waste Management System:** Monitors and categorizes collected plastic waste, enabling efficient processing and ensuring a steady supply of raw materials for eco-lumber production.
2. **Recycling Equipment Management:** Tracks the status and performance of recycling machinery to prevent downtime and maintain efficiency.
3. **Community Engagement Platform:** Facilitates communication with communities, tracks participation in workshops, and collects feedback to improve outreach efforts.
4. **Data Analytics Tools:** Analyzes operational metrics, financial performance, and community impact to support actionable decision-making.
5. **Training Management System:** Manages employee training programs, skill development, and monitors workforce adaptation to new technologies and practices.

By leveraging these data types and systems, PLAF efficiently manages its operations, enhances community engagement, and drives its mission of mitigating the environmental and social impacts of plastic waste.

I. EA Management Plan

EA Program Management

The Enterprise Architecture (EA) Management Plan for Plastic Flamingo (PLAF) aligns the organization's sustainability goals with its operational and technological framework. It provides a structured guide to manage, implement, and assess EA initiatives, ensuring efficiency, scalability, and sustainability.

1.1 Governance and Principles

The first step in the EA Management Plan is defining governance and principles. PLAF's governance framework ensures alignment of EA initiatives with the organization's mission to combat plastic pollution and promote sustainability. The principles guiding this framework are alignment of EA projects with environmental and operational goals, transparency in communication and decision-making, and consistency through standardized systems and processes. These principles ensure that the EA initiatives effectively support PLAF's objectives.

1.2 Support for Strategy and Business

The EA program serves as a strategic enabler by leveraging data-driven decisions to improve recycling rates and community programs, enhancing operational efficiency from plastic collection to eco-lumber production, and fostering innovation by adopting advanced recycling and digital technologies. This step ensures that EA initiatives contribute to the organization's long-term sustainability goals and growth.

1.3 EA Roles and Responsibilities

EA initiatives are guided by clear roles to ensure accountability and progress. The EA Lead oversees architecture development, ensuring alignment with organizational goals. The Technology Manager handles systems and technological operations, while the operations coordinator connects technical teams with field operations. Stakeholders provide critical input to align initiatives with community and partner needs, ensuring a collaborative approach.

1.4 EA Program Budget

PLAF's EA program prioritizes investments in technological enhancements, such as recycling systems and analytics tools, infrastructure improvements like cloud platforms and machinery upgrades, and employee training programs for eco-friendly practices and new technologies. Monitoring the budget ensures cost-effectiveness and alignment with the organization's priorities, making the best use of available resources

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1.5 EA Program Performance Measures

PLAF focuses on maintaining detailed documentation of system changes, using structured evaluation processes to assess updates, and ensuring compatibility across evolving systems through version control. These practices ensure effective management of architectural changes and sustain the integrity of the EA framework.

Summary of Current Architecture

The existing architecture supports PLAF's goals but has room for upgrades to improve efficiency and scalability.

2.1 Strategic Goals and Initiatives

PLAF aims to reduce plastic waste through improved collection and recycling methods, increase the production and distribution of eco-lumber, and strengthen community sustainability efforts.

PLAF has launched several initiatives, including the adoption of advanced recycling technologies, the development of digital platforms for data management and community engagement, and the expansion of partnerships with both local and global organizations. Through these efforts, PLAF aims to significantly impact environmental sustainability and foster a more resilient and eco-friendly future.

2.2 Business Services and Information Flows

PLAF provides essential services focused on plastic collection, recycling, and community outreach. They rely on key information flows to enhance their operations and impact. Community data is collected from workshop feedback, which is used to refine and improve their programs. Operational data monitors waste volumes and recycling outputs to ensure efficient operations and maintain high-quality standards. Financial data tracks financial metrics to ensure sustainability and optimize resource use. By leveraging these data streams, PLAF can continually improve their services and contribute more effectively to environmental sustainability.

2.3 Systems and Applications

PLAF currently utilizes a Waste Management System to track and organize plastic waste effectively. Additionally, they use Engagement Platforms to connect with communities and gather valuable feedback. These systems help PLAF streamline their operations and improve their overall impact on sustainability.

2.4 Technology Infrastructure

PLAF's infrastructure includes stable internet connectivity for cloud systems and communication, recycling equipment equipped with integrated monitoring tools, and local networks that enable real-time data sharing. These components ensure efficient and seamless operations across their various initiatives.

2.5 IT Security

PLAF implements several security measures to safeguard its operations. These include role-based access controls to protect sensitive data, firewalls and antivirus tools to ensure cyber security, and regular data backups to prevent data loss. These measures are essential for maintaining the integrity and security of PLAF's systems and information.

Summary of Future Architecture

Future architecture emphasizes scalability, efficiency, and sustainability.

3.1 Future Operating Scenarios

PLAF plans to enhance its operations by implementing automated sorting and recycling processes to improve efficiency. Additionally, they aim to develop mobile-friendly platforms for better community outreach and engagement. Furthermore, PLAF will leverage predictive analytics to forecast recycling trends and material needs, allowing for more accurate planning and resource allocation. These initiatives will help PLAF to streamline operations and expand their impact on sustainability.

3.2 Planning Assumptions

PLAF's future plans assume the availability of reliable internet for cloud systems, increased funding for technology upgrades, and comprehensive employee training for advanced tools. These elements are essential to support the organization's growth and enhance its operational efficiency.

3.3 Updating Current and Future EA Views

Continuous improvements at PLAF involve addressing inefficiencies by implementing better tools, defining a target architecture that incorporates automation and AI-driven solutions, and conducting regular reviews to ensure alignment with organizational goals. These efforts aim to enhance operational efficiency and support the achievement of PLAF's mission.

3.4 EA Sequencing Plan

Key priorities for PLAF include upgrading waste management systems to enhance efficiency, introducing cloud-based analytics tools for better data management, developing mobile platforms to improve community engagement, and scaling operations with automated recycling

technology to handle increased volumes of plastic waste. These initiatives aim to strengthen PLAF's capabilities and extend its impact on environmental sustainability.

3.5 EA Configuration Management

To manage changes effectively, PLAF emphasizes thorough documentation, ensuring detailed records of all system changes. They implement structured change management processes to evaluate and oversee updates systematically. Additionally, they use version control to maintain compatibility across updated systems, ensuring smooth integration and operation. These practices help PLAF adapt to changes while maintaining stability and efficiency in their operations.

EA Glossary

1. Eco-lumber:

An eco-friendly building material obtained from recycled plastics, which has the potential to replace natural wood. It is utilized in construction projects such as outdoor furniture, decking, and fencing.

2. Circular Economy:

This refers to the concept of an economic system that ensures minimal waste and maximum usage of resources. It involves reusing, repairing, refurbishing, and recycling materials in place of the traditional linear economy, which follows the pattern of "take, make, dispose".

3. PLAF (Plastic Flamingo):

A social enterprise from the Philippines, 2018, it was focused on reducing plastic pollution through collection of plastic waste that gets recycled to be used as eco-lumber and other sustainable products.

4. Waste Management System:

A tracking system designed for the proper management of plastic wastes collected that enables smooth recycling and an adequate supply of raw materials needed in producing the material for making the eco-lumber.

5. Recycling Equipment Management:

The process of monitoring the performance and status of machinery used in the recycling process to prevent downtime and ensure efficiency.

6. Community Engagement Data:

Data collected from community members who participate in workshops, plastic collection programs, and feedback surveys to help PLAF improve its programs and measure their impact.

7. Data on Recycling:

This will be data concerning the quantity of plastic collected, recycling percentages, and quantity of eco-lumber produced. Such information helps track performance concerning the operation as well as quality.

8. Supply Chain Data:

Concerning the supply chain, this will involve data regarding partnerships with local governments, logistics, and suppliers that assist PLAF in effectively managing material flow from collection to recycling.

9. Data Analytics Tools:

Software and systems that analyze operational data, financial performance, and community impact to give insights for decision-making and process optimization.

10. Technology Infrastructure:

The physical and digital resources needed for PLAF's operations, including internet connectivity, local area networks (LAN), recycling machinery, and communication tools.

11. EA-Enterprise Architecture:

An organized set of principles and guidelines, which an organization applies for the alignment of its business strategies with the technological structure. In the case of PLAF, EA ensures that the operational sustainability goals of the organization are supported through its technological framework and system.

12. Stakeholders:

Individuals or groups that have an interest in PLAF's operations and outputs, such as local government, community members, businesspeople, and international bodies.

13. Governance Framework:

The rules, practices, and processes followed to govern and control an organization's operations so that it meets its goals and objectives.

14. IT Security:

This involves protection of PLAF's digital infrastructure and data by enforcing role-based access controls, installing firewalls, antivirus tools, and having regular data backup in order to prevent data loss or cyber-attacks.

15. Scalability:

The capacity of PLAF's systems, operations, and technologies to scale up and scale out to accommodate increased demand or expanded operations, such as expansion to new regions.

16. Predictive Analytics:

Techniques that use data, statistical algorithms, and machine learning to differentiate the likelihood of future outcomes based on historical data. PLAF intends to apply this technology in forecasting recycling trends and material needs.

References

[1] Plastic Flamingo, "Plastic Flamingo: Reducing Plastic Waste and Promoting Sustainability," [Online]. Available: <https://www.plasticflamingo.org/>. [Accessed: Dec. 15, 2024].